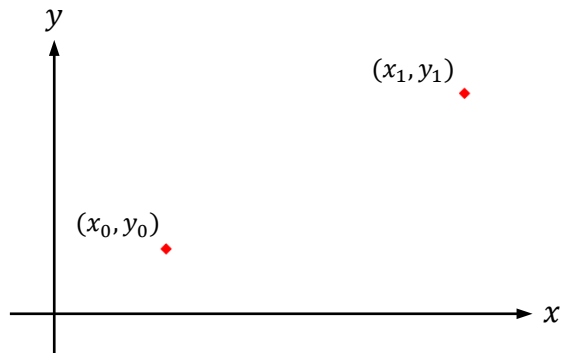


Lagrange Interpolating Polynomials



Linear Interpolation

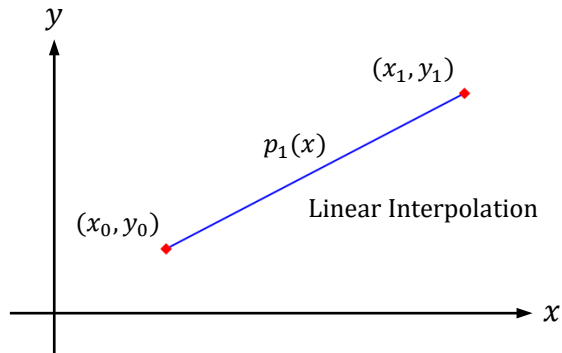
$$p_1(x) = L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1); \quad x_0 \leq x \leq x_1$$

$$L_0^{(1)}(x) = a_0x + b_0: \quad \left. \begin{array}{l} L_0^{(1)}(x_0) = 1 \rightarrow 1 = a_0 \cdot x_0 + b_0 \\ L_0^{(1)}(x_1) = 0 \rightarrow 0 = a_0 \cdot x_1 + b_0 \end{array} \right\} \begin{array}{l} a_0 = \frac{1}{x_0 - x_1} \\ b_0 = \frac{-x_1}{x_0 - x_1} \end{array}$$

$$L_0^{(1)}(x) = \frac{x - x_1}{x_0 - x_1}$$

$$L_1^{(1)}(x) = a_1x + b_1: \quad \left. \begin{array}{l} L_1^{(1)}(x_0) = 0 \rightarrow 0 = a_1 \cdot x_0 + b_1 \\ L_1^{(1)}(x_1) = 1 \rightarrow 1 = a_1 \cdot x_1 + b_1 \end{array} \right\} \begin{array}{l} a_1 = \frac{1}{x_1 - x_0} \\ b_1 = \frac{-x_0}{x_1 - x_0} \end{array}$$

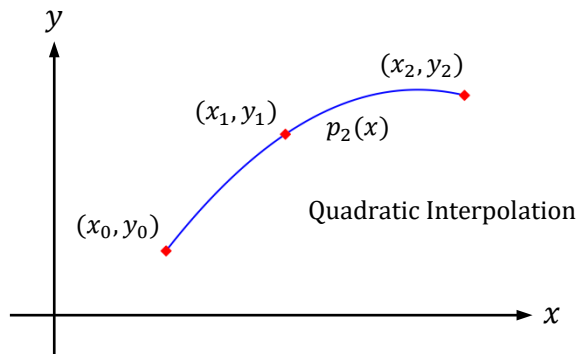
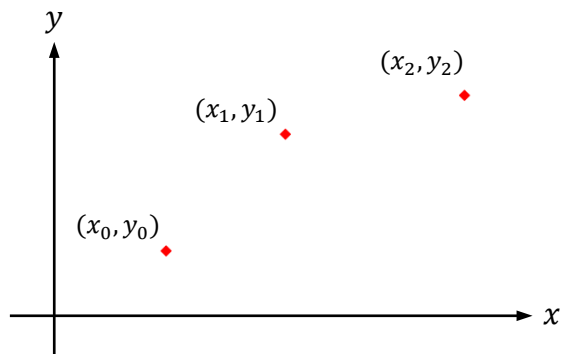
$$L_1^{(1)}(x) = \frac{x - x_0}{x_1 - x_0}$$



For $x_0 \leq x \leq x_1$,

$$p_1(x) = \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1)$$

Lagrange Interpolating Polynomials



Quadratic Interpolation

$$p_2(x) = L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2); \quad x_0 \leq x \leq x_2$$

$$L_0^{(2)}(x) = a_0x^2 + b_0x + c_0: \quad \left. \begin{array}{l} L_0^{(2)}(x_0) = 1 \rightarrow 1 = a_0 \cdot x_0^2 + b_0 \cdot x_0 + c_0 \\ L_0^{(2)}(x_1) = 0 \rightarrow 0 = a_0 \cdot x_1^2 + b_0 \cdot x_1 + c_0 \\ L_0^{(2)}(x_2) = 0 \rightarrow 0 = a_0 \cdot x_2^2 + b_0 \cdot x_2 + c_0 \end{array} \right\} \begin{array}{l} a_0 = \frac{1}{(x_0-x_1) \cdot (x_0-x_2)} \\ b_0 = \frac{-x_1-x_2}{(x_0-x_1) \cdot (x_0-x_2)} \\ c_0 = \frac{x_1 \cdot x_2}{(x_0-x_1) \cdot (x_0-x_2)} \end{array}$$

$$L_0^{(2)}(x) = \frac{x^2 - (x_1 + x_2) \cdot x + x_1 \cdot x_2}{(x_0 - x_1) \cdot (x_0 - x_2)} = \frac{(x - x_1) \cdot (x - x_2)}{(x_0 - x_1) \cdot (x_0 - x_2)}$$

$$L_1^{(2)}(x) = a_1x^2 + b_1x + c_1: \quad L_1^{(2)}(x) = \frac{x^2 - (x_0 + x_2) \cdot x + x_0 \cdot x_2}{(x_1 - x_0) \cdot (x_1 - x_2)} = \frac{(x - x_0) \cdot (x - x_2)}{(x_1 - x_0) \cdot (x_1 - x_2)}$$

$$L_2^{(2)}(x) = a_2x^2 + b_2x + c_2: \quad L_2^{(2)}(x) = \frac{x^2 - (x_0 + x_1) \cdot x + x_0 \cdot x_1}{(x_2 - x_0) \cdot (x_2 - x_1)} = \frac{(x - x_0) \cdot (x - x_1)}{(x_2 - x_0) \cdot (x_2 - x_1)}$$

For $x_0 \leq x \leq x_2$,

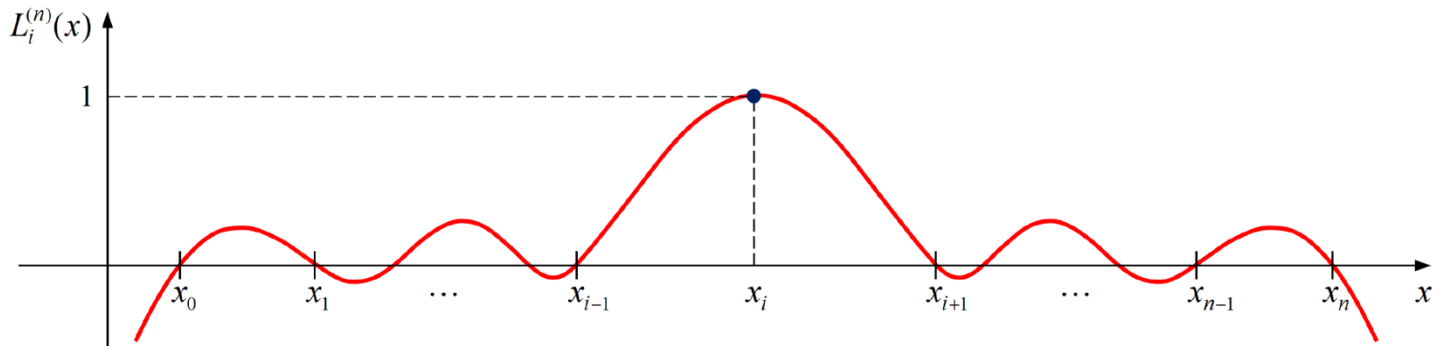
$$p_2(x) = \frac{(x - x_1) \cdot (x - x_2)}{(x_0 - x_1) \cdot (x_0 - x_2)} \cdot f(x_0) + \frac{(x - x_0) \cdot (x - x_2)}{(x_1 - x_0) \cdot (x_1 - x_2)} \cdot f(x_1) + \frac{(x - x_0) \cdot (x - x_1)}{(x_2 - x_0) \cdot (x_2 - x_1)} \cdot f(x_2)$$

Lagrange Interpolating Polynomials

General Form

$$p_n(x) = \sum_{i=0}^n L_i^{(n)}(x) \cdot f(x_i); \quad x_0 \leq x \leq x_n$$

$$L_i^{(n)}(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \quad \rightarrow \quad L_i^{(n)}(x_j) = \begin{cases} 1; & i = j \\ 0; & i \neq j \end{cases}$$



Lagrange Interpolating Polynomials

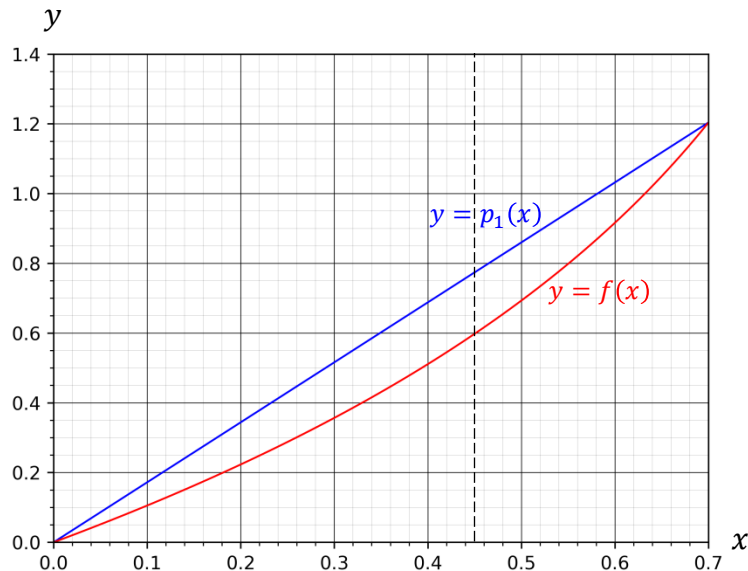
Example 04-02a: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728

$$\begin{aligned}
 p_1(x) &= L_0^{(1)}(x) \cdot f(x_0) + L_1^{(1)}(x) \cdot f(x_1) \\
 &= \frac{x - x_1}{x_0 - x_1} \cdot f(x_0) + \frac{x - x_0}{x_1 - x_0} \cdot f(x_1) \\
 f(0.45) &\approx p_1(0.45) = \frac{0.45 - 0.7}{0.0 - 0.7} \cdot 0.0000000 + \frac{0.45 - 0.0}{0.7 - 0.0} \cdot 1.2039728 \\
 &= 0.7739825
 \end{aligned}$$

$$f(x) = -\ln(1 - x)$$

$$f(0.45) = -\ln(1 - 0.45) = 0.5978370$$



Lagrange Interpolating Polynomials

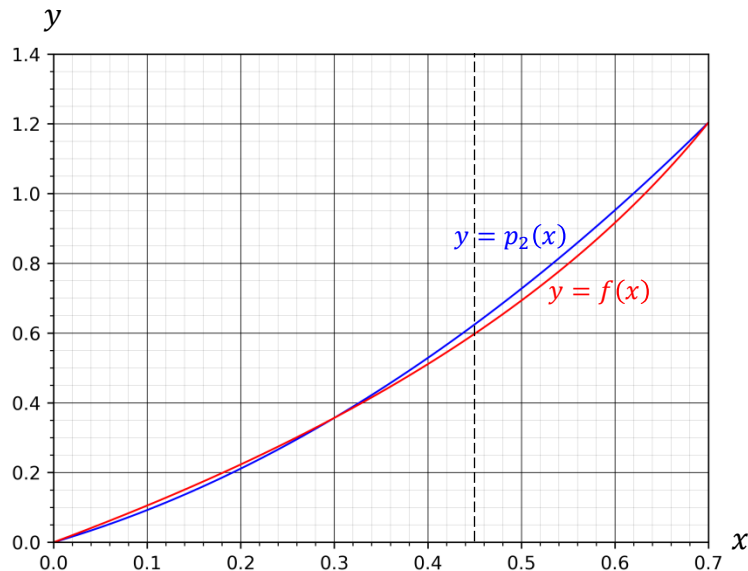
Example 04-02b: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749

$$\begin{aligned}
 p_2(x) &= L_0^{(2)}(x) \cdot f(x_0) + L_1^{(2)}(x) \cdot f(x_1) + L_2^{(2)}(x) \cdot f(x_2) \\
 &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \cdot f(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \cdot f(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \cdot f(x_2) \\
 f(0.45) &\approx p_2(0.45) = \frac{(0.45-0.7)(0.45-0.3)}{(0.0-0.7)(0.0-0.3)} \cdot 0.0000000 + \frac{(0.45-0.0)(0.45-0.3)}{(0.7-0.0)(0.7-0.3)} \cdot 1.2039728 \\
 &\quad + \frac{(0.45-0.0)(0.45-0.7)}{(0.3-0.0)(0.3-0.7)} \cdot 0.3566749 \\
 &= 0.6246262
 \end{aligned}$$

$$f(x) = -\ln(1-x)$$

$$f(0.45) = -\ln(1-0.45) = 0.5978370$$



Lagrange Interpolating Polynomials

Example 04-02c: Approximate $f(0.45)$ from the following data using Lagrange polynomials.

i	x_i	$f(x_i)$
0	0.0	0.0000000
1	0.7	1.2039728
2	0.3	0.3566749
3	0.2	0.2231436

$$p_3(x) = L_0^{(3)}(x) \cdot f(x_0) + L_1^{(3)}(x) \cdot f(x_1) + L_2^{(3)}(x) \cdot f(x_2) + L_3^{(3)}(x) \cdot f(x_3)$$

$$\begin{aligned}
 f(0.45) \approx p_3(0.45) &= \frac{(0.45-0.7) \cdot (0.45-0.3) \cdot (0.45-0.2)}{(0.0-0.7) \cdot (0.0-0.3) \cdot (0.0-0.2)} \cdot 0.0000000 \\
 &+ \frac{(0.45-0.0) \cdot (0.45-0.3) \cdot (0.45-0.2)}{(0.7-0.0) \cdot (0.7-0.3) \cdot (0.7-0.2)} \cdot 1.2039728 \\
 &+ \frac{(0.45-0.0) \cdot (0.45-0.7) \cdot (0.45-0.2)}{(0.3-0.0) \cdot (0.3-0.7) \cdot (0.3-0.2)} \cdot 0.3566749 \\
 &+ \frac{(0.45-0.0) \cdot (0.45-0.7) \cdot (0.45-0.3)}{(0.2-0.0) \cdot (0.2-0.7) \cdot (0.2-0.3)} \cdot 0.2231436 \\
 &= 0.6045237
 \end{aligned}$$

$$f(x) = -\ln(1-x)$$

$$f(0.45) = -\ln(1-0.45) = 0.5978370$$

